



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-Section SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: User Registration and Authentication

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1. Introduction

1.1. Purpose

This Functional Requirements Specification (FRS) defines the functional and non-functional requirements for the system's User Registration and Authentication component. It serves as the baseline for design, implementation, and verification for stakeholders, analysts, developers, and testers.

1.2. Scope

The scope of this system is strictly limited to user account management and access control. The system will encompass the following core capabilities:

- User account registration.
- User login and session establishment.
- User logout and session termination.
- Validation and authorization to ensure protected pages are inaccessible without successful authentication.
- Allowing authenticated users to view their profile or dashboard.

This system does not include features outside of authentication and authorization, such as payment processing, data analysis, or other application-specific business logic.

1.3. Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
FRS	Functional Requirements Specification
User	A person who interacts with the system.
Guest User	An unauthenticated User who has access only to public sections and registration/login forms.
Authenticated User	A User who has successfully logged in and has access to all authorized pages, including protected sections and their profile/dashboard.
UI	User Interface. The React-based frontend application.
API	Application Programming Interface. The Spring Boot backend service.
ERD	Entity-Relationship Diagram

2. Overall Description

2.1. System Perspective

The User Registration and Authentication system is a foundational security module integrated within a larger application architecture. It acts as the gatekeeper for access to protected resources. The system receives requests from the **React UI** (client), processes the

authentication and authorization logic within the **Spring Boot API** (server), and manages user data persistence in the **Database**.

2.2. User Classes and Characteristics

User Class	Characteristics	Permissions
Guest User	Not currently logged into the system.	Can access public pages. Can initiate the Register and Login processes. Cannot access protected pages.
Authenticated User	Successfully logged into the system and possesses a valid session/token.	Can access all public and protected pages. Can View Profile/Dashboard and Log Out.

2.3. Operating Environment

Component	Technology / Platform
Frontend	React UI (Web Browser compatibility assumed)
Backend / API	Spring Boot (Java) running on a server environment.
Database	Relational Database Management System (e.g., MySQL, PostgreSQL).
Communication	RESTful API calls over HTTPS.

2.4. Assumptions and Dependencies

Type	Item	Description
Assumption	User Input Validity	It is assumed that the UI performs initial client-side validation on user input (e.g., required fields, email format) before submission to the API.
Assumption	External Services	The system assumes the availability and functionality of necessary infrastructure (e.g., DNS, network, load balancers).
Dependency	Database Schema	Successful operation is dependent on the users table existing and being correctly structured as defined in the ERD.
Dependency	Secure Communication	The deployment environment must support and enforce HTTPS for all client-server communication.

3. System Features and Functional Requirements

3.1. Feature 1: Account Registration

Description: Allows a guest user to create a new account by providing the needed account information and requirements

Functional Requirements:

- The system should present a registration form requiring a unique identifier and a password.
- The system should check that all mandatory fields are filled and meet the defined format rules.
- The system will verify that the identifier provided is unique and not used by another user.
- Upon successful validation, the system will securely hash and salt the user's password before storing the new User record in the Database.
- The system will return a success confirmation when an account is created

3.2. Feature 2: User Authentication and Management

Description: Manages the user login, maintains the user's authenticated session, allows access to protected resources, and handles session termination

Functional Requirements:

- The system will accept a User's identifier and password for authentication.
- The system should securely verify the provided password against the stored hash in the Database.
- Upon successful authentication, the system will generate a secure authentication token and return it to the UI for session management.
- The system should deny access to any protected page or API endpoint if the request does not contain a valid, unexpired authentication token.
- The system will allow an Authenticated User to View their Profile/Dashboard.
- The system should invalidate the user's active session/token upon a Log Out request, effectively reverting the user's status to a Guest User.

4. Non-Functional Requirements

Security

- All user passwords should be stored using a strong, industry-standard hashing and salting algorithm.
- All communication between the UI and the API will be encrypted using HTTPS/TLS.
- Authentication tokens (e.g., JWT) will be managed securely and include appropriate expiration limits.
- The system should implement rate limiting on login attempts to mitigate brute-force attacks.

Performance

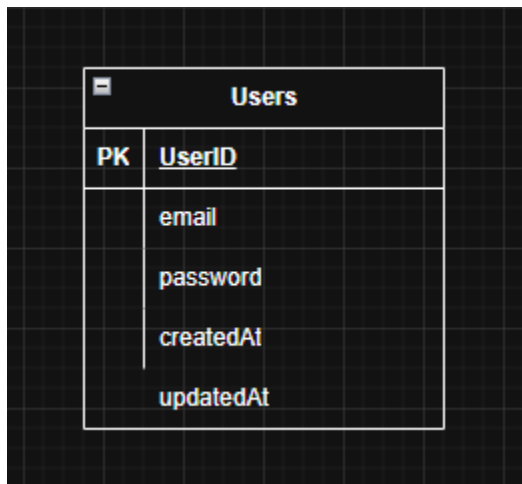
- The average response time for a **Login** or **Registration** request will be under 1.5 seconds under normal load conditions.

Reliability and Availability

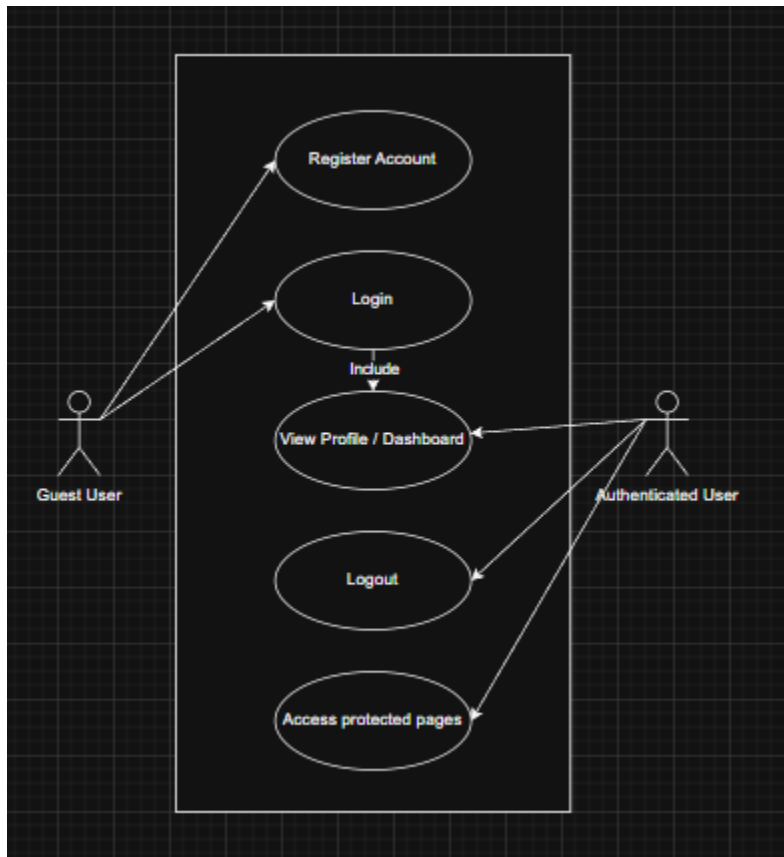
- The Authentication Service should maintain an availability of 99.9% during peak operating hours.
- The system will provide clear and informative error messages to the user in case of authentication or registration failure.

5. System Models (Diagrams)

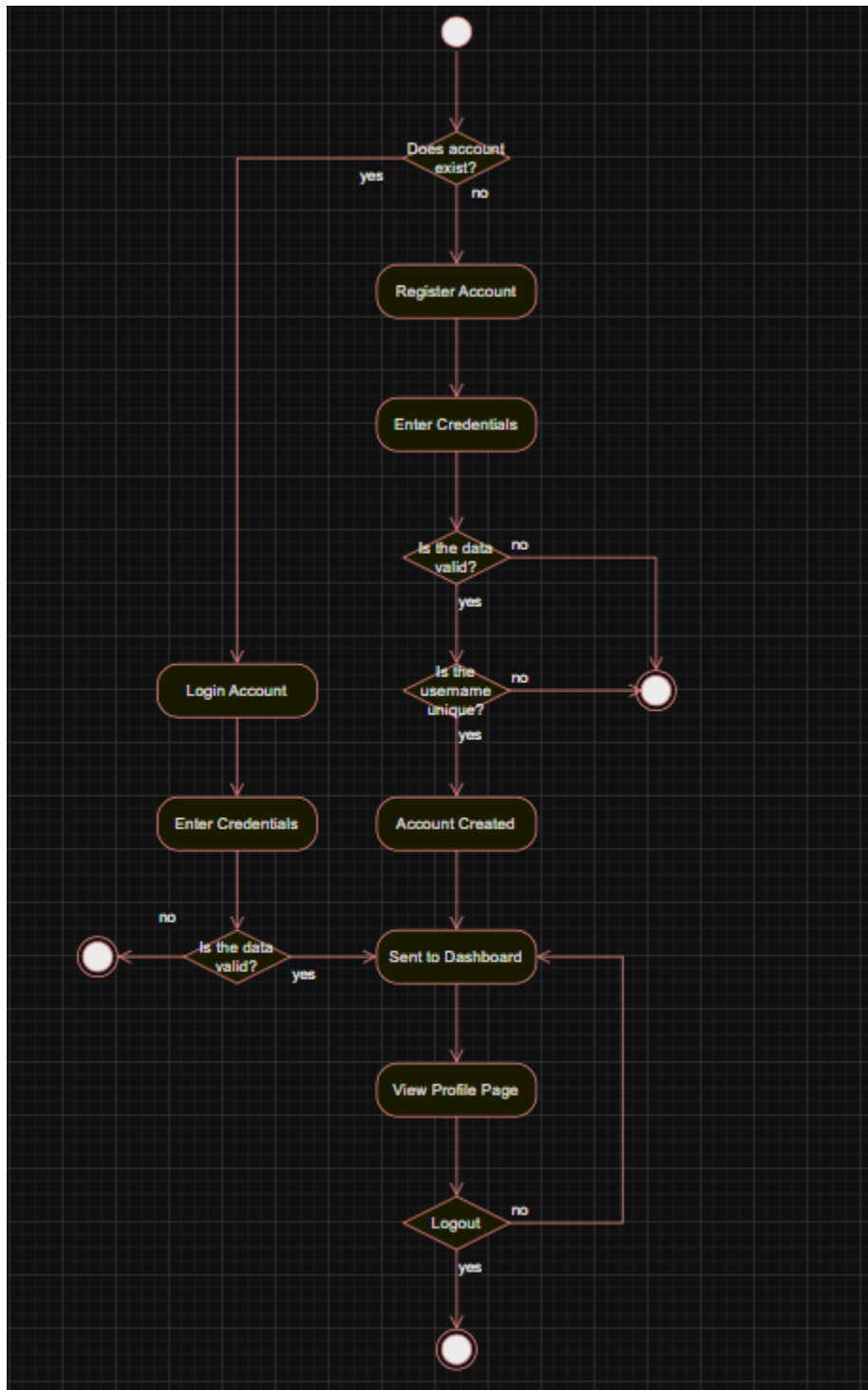
5.1. ERD



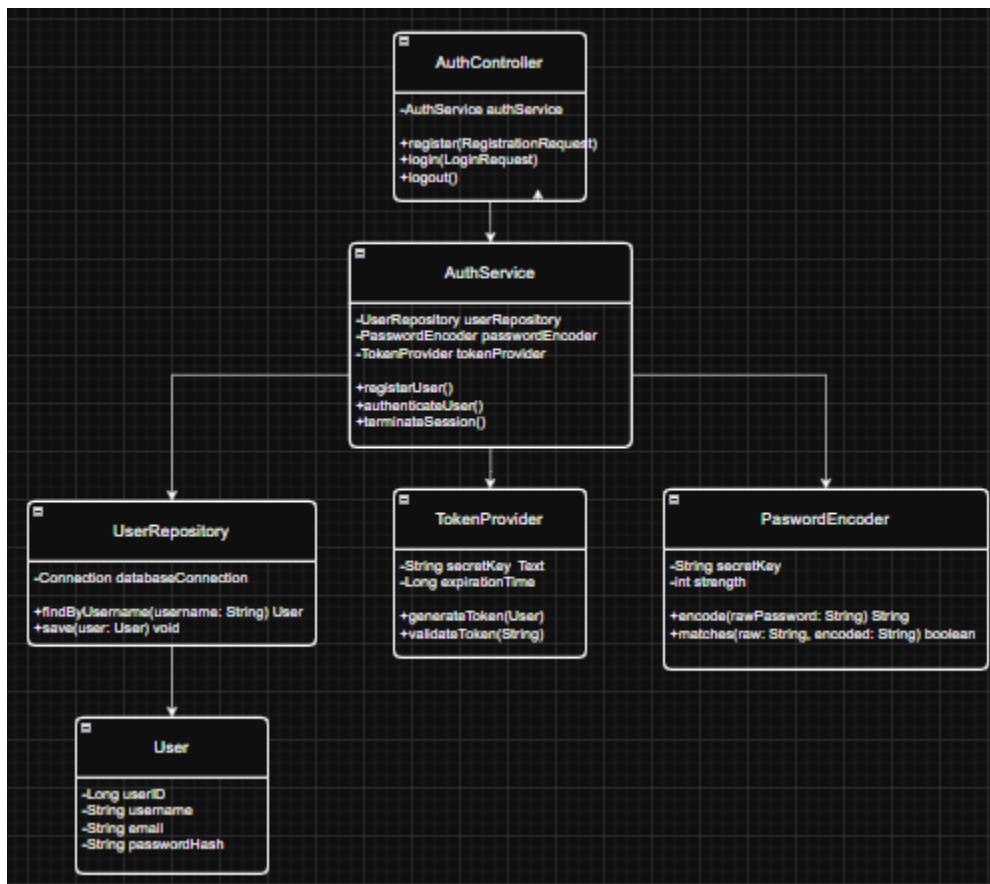
5.2. Use Case Diagram



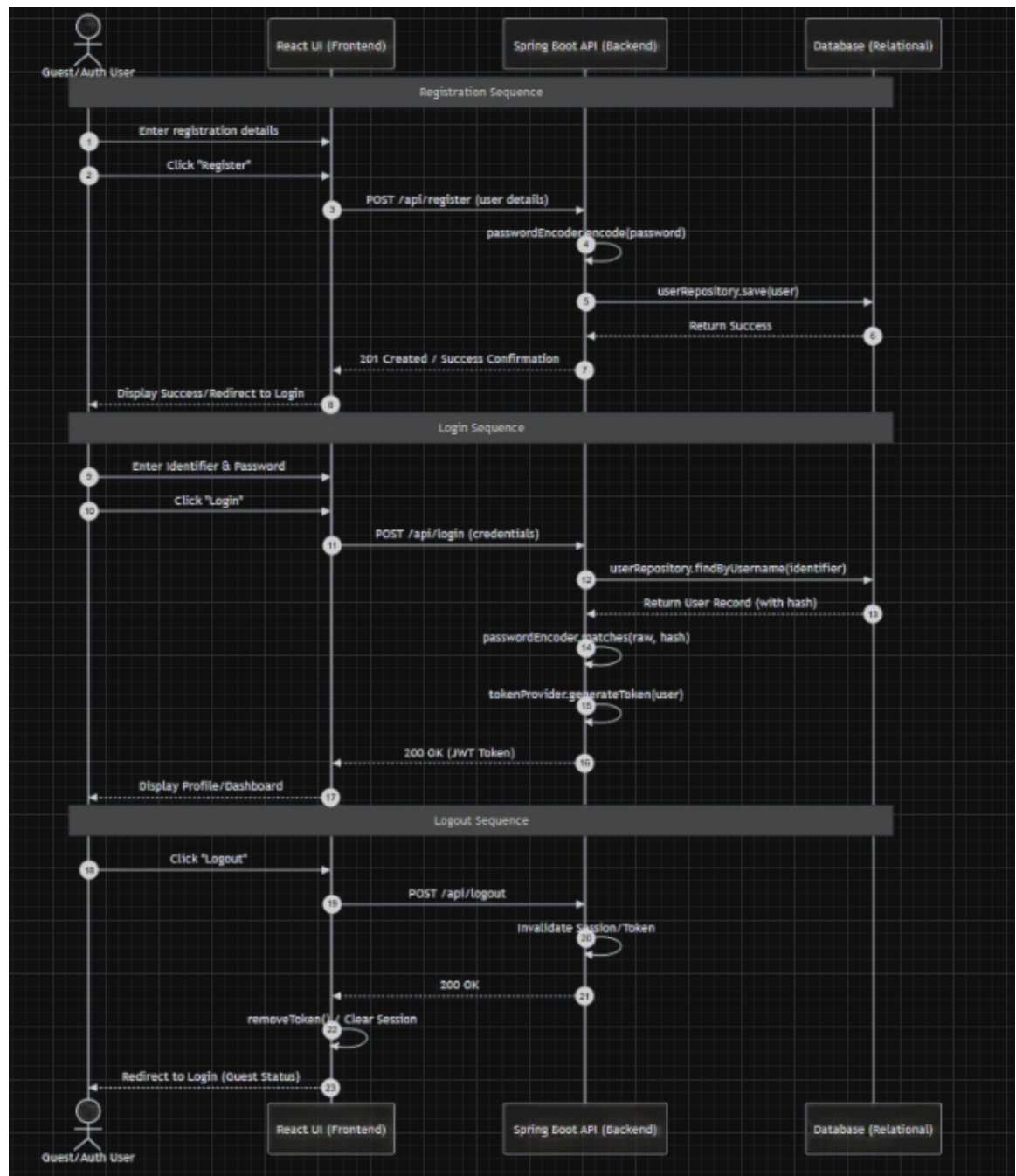
5.3. Activity Diagram



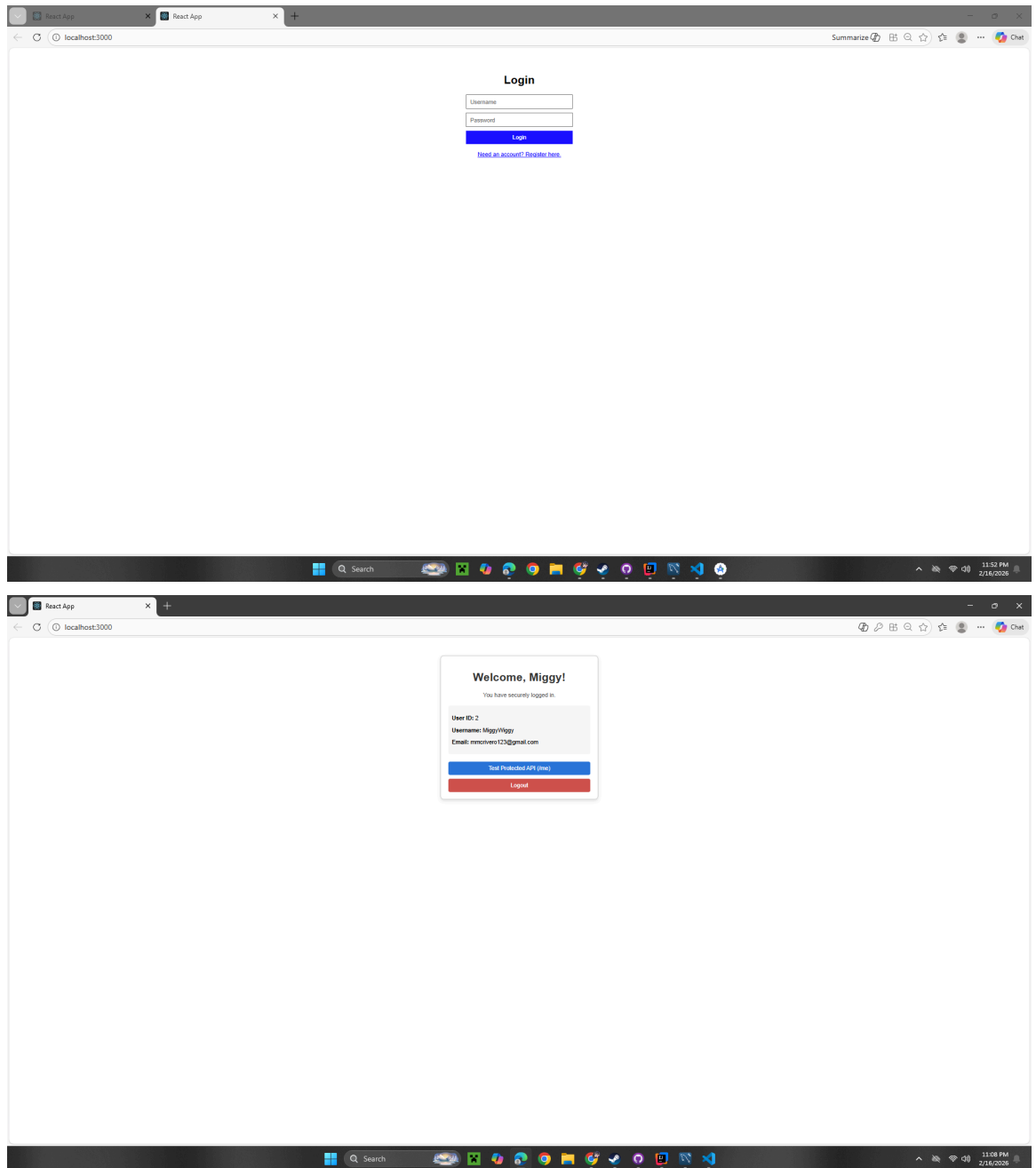
5.4. Class Diagram



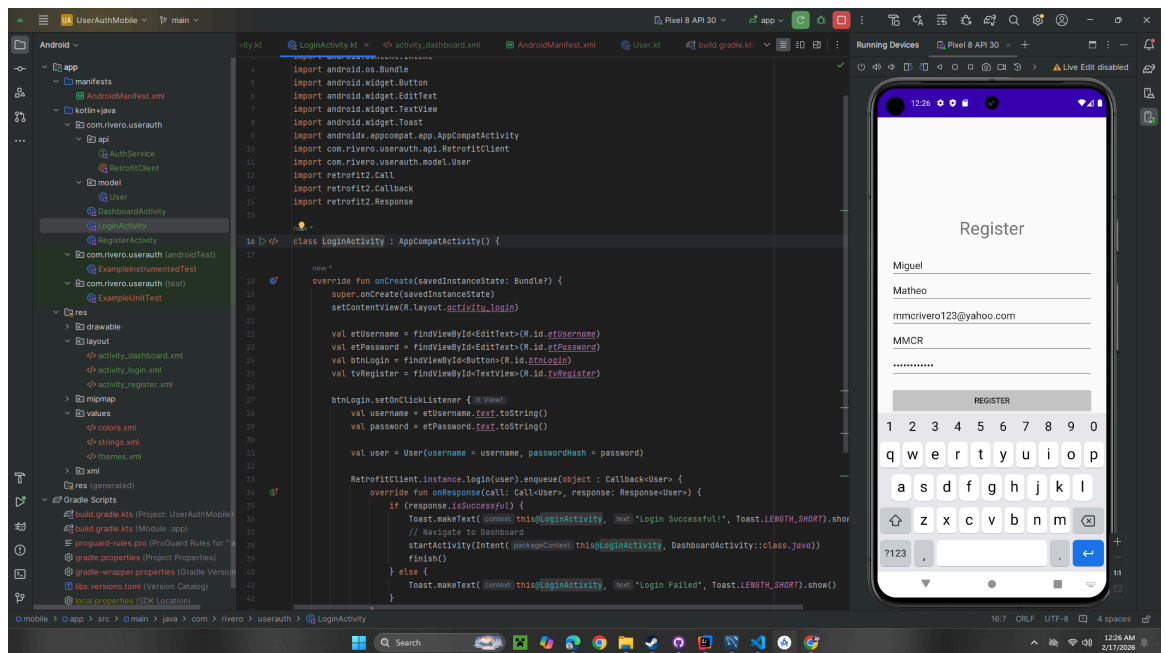
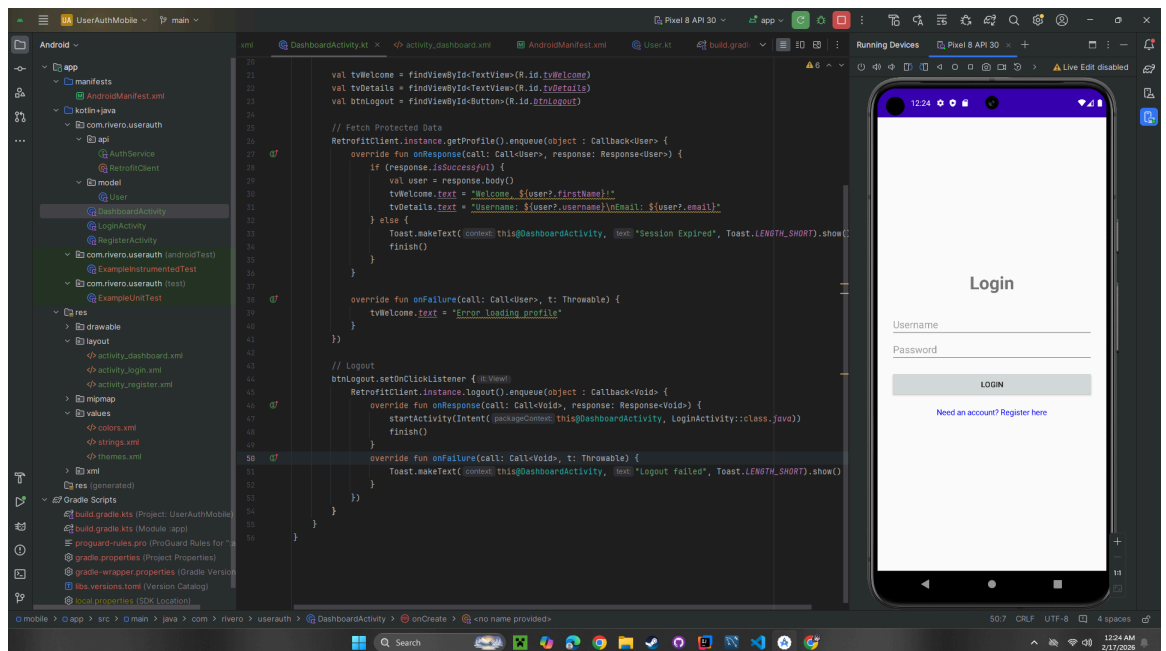
5.5. Sequence Diagram



5.6. Front End and Dashboard



5.7. Mobile Login and Register



6. Appendices

Include any additional information, references, or support materials.